

Course Content	Unit 1: Fundamentals of Artificial Intelligence and Data Science 1.1 Introduction to Artificial Intelligence (AI) 1.1.1 Definition and history of AI 1.1.2 Key concepts and terminologies 1.1.3 Applications of AI in various fields 1.2 Introduction to Data Science 1.2.1 Definition and scope of Data Science 1.2.2 Data Science vs. AI 1.2.3 Role of a Data Scientist 1.2.4 Basic Terminologies 1.2.4.1 Data, Information, and Knowledge 1.2.4.2 Structured Data, Unstructured Data Unit 2: Data Analytics and Business Intelligence 2.1 Introduction to Data Analytics 2.1.1 Types of Data Analytics: 2.1.1.1 Descriptive, Diagnostic, Predictive, and Prescriptive 2.1.2 Data Analytics Process and Life Cycle 2.1.3 Tools and Technologies for Data Analytics 2.1.4 Business Intelligence (BI) 2.1.4.1 Definition and Importance of BI 2.1.4.2 Components of BI Systems 2.1.4.3 BI vs. Data Analytics 2.2 Data Visualization 2.2.1 Importance of Data Visualization 2.2.2 Tools for Data Visualization (e.g., Tableau, Power BI) 2.2.3 Principles of Effective Data Visualization Unit 3: Machine Learning and A.I. Technique Basics 3.1 Introduction to Machine Learning (ML) 3.2 Definition and types of Machine Learning 3.2.1 Supervised, Unsupervised, and Reinforcement Learning 3.3 Advanced AI Techniques Terminologies 3.3.1 Natural Language Processing (NLP) 3.3.2 Computer Vision 3.3.3 AI in Robotics Unit 4 : Emerging Technologies and Ethical Considerations 4.1 Emerging Technologies in AI and Data Science 4.2 Overview of Big Data technologies (e.g., Hadoop, Spark) 4.3 Internet of Things (IoT) and its connection with AI 4.4 Cloud Computing in AI and Data Science Unit 5 : Ethical and Societal Implications of A.I. 5.1 Ethical issues in AI and Data Science (bias, privacy, and security) 5.2 Responsible AI practices 5.3 Impact of AI on jobs and society 5.4 Future Trends and Career Opportunities 5.5 Current trends and future directions in AI and Data Science 5.6 Skills and competencies required for careers in AI and Data Analytics 5.7 Job roles and career paths in the field
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Reference Books	1. "Artificial Intelligence: A Modern Approach" by Stuart Russell, Peter Norvig, Pearson, ISBN: 978-9332582387 2. "Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking" by Foster Provost, Tom Fawcett, O'Reilly Media, ISBN: 978-1449361327 3. "Python for Data Analysis" by Wes McKinney, O'Reilly Media, ISBN: 978-1491957660 4. "Data Mining: Concepts and Techniques" by Jiawei Han, Micheline Kamber, Jian Pei, Morgan Kaufmann, ISBN: 978-0123814791 5. "Machine Learning: A Probabilistic Perspective" by Kevin P. Murphy, The MIT Press, ISBN: 978-0262018029 6. "Artificial Intelligence: Foundations of Computational Agents" by David L. Poole, Alan K. Mackworth, Cambridge University Press, ISBN: 978-1107195394 (Indian Edition) 7. "Data Science for Dummies" by Lillian Pierson, John Wiley & Sons, ISBN: 978-1119327639 (Indian Edition) 8. "Python Data Science Handbook" by Jake VanderPlas, O'Reilly Media, ISBN: 978-1491912058 (Indian Edition) 9. "Data Mining Techniques: For Marketing, Sales, and Customer Relationship Management" by Gordon S. Linoff, Michael J. A. Berry, Wiley India Pvt Ltd, ISBN: 978-8126514366 10. "Introduction to Machine Learning with Python: A Guide for Data Scientists" by Andreas C. Müller, Sarah Guido, O'Reilly Media, ISBN: 978-1449369415 (Indian Edition)
Teaching Methodology	Class Work, Discussion, Self-Study, Seminars and/or Assignments
Evaluation Method	50% Internal assessment. 50% External assessment.